
COMPUTER AIDED ENGINEERING DRAWING LABORATORY

Course Code: MELC28

Credits: 0:1:1

Pre-requisites: -

Contact Hours: 14T+14P

Course Coordinator: Dr. Mohandas K N

Course Contents

Course Learning Objectives:

1. To understand the basic and applied engineering drawings solutions by using suitable software.
2. To comprehend the criticality of development of typical lateral surfaces.
3. To visualize typical engineering components.

Teaching-Learning (General Instructions):

- Students should be made aware of powerful engineering communication tool – Drawing.
- Simple Case studies can be suitably selected by the teacher for hands on practice to induce the feel of fruitfulness of learning.
- Appropriate Models, Power Point presentation, Charts, Videos, shall be used to enhance visualization before hands on practice.
- For application problems use very generally available actual objects. (Example: For rectangular prism / object; matchbox, carton boxes, book, etc can be used. Similarly for other shapes).
- Use any CAD software for generating orthographic and pictorial views.
- Make use of sketch book with graph sheets for manual / preparatory sketching.

Unit I

Introduction:

Significance of Engineering drawing, BIS Conventions of Engineering Drawing, Free hand sketching of engineering drawing, Scales. Introduction to Computer Aided Drafting software, Co-ordinate system and reference planes HP, VP, RPP & LPP of 2D environment. Commands and creation of Lines, coordinate points, axes, polylines, square, rectangle, polygons, splines, circles, ellipse, text, move, copy, off-set, mirror, rotate, trim, extend, break, chamfer, fillet and curves.

Orthographic Projections of Points, Lines and Planes:

Orthographic projections: Definitions - Planes of projection, reference line and conventions employed, Projections of points in First and Third quadrants (**No problems**), Projections of straight lines (located in First quadrant/first angle only), True and apparent lengths, True and apparent inclinations to reference planes (**No problems**). Orthographic Projections of Planes viz triangle, square, rectangle, pentagon, hexagon, and circular laminae (**Placed in First quadrant only using change of position method**).

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Orthographic Projections: <https://youtu.be/uePTMVQIIA4>
- Links: Orthographic projections of plane: <https://www.youtube.com/watch?v=GguSUMNxc8Q>

Unit II

Development of Lateral Surfaces of Solids:

Development of sectioned (**Section plane perpendicular to VP and inclined to HP bisecting the axis only**) lateral surfaces of right regular prism, cylinder, pyramid and cone resting **with base on HP only**.

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Development of lateral surfaces: https://www.youtube.com/watch?v=U5mz9_W-xdI

Unit III

Orthographic Projection of Solids:

Orthographic projection of right regular solids (**Solids Resting on HP only**): Prism & Pyramid (triangle, square, rectangle, pentagon, hexagon), Cylinder, Cone and Cube (No freely suspended problems).

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Orthographic Projection of Solids: <https://youtube.com/-sUEGrynfJ0>

Unit IV

Isometric Projections:

Isometric scale, Isometric projection of hexahedron (cube), right regular prism, pyramid, cylinder, cone, sphere and frustum of solid. Isometric projection of combination of two simple solids (**Co-axial only**).

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Isometric Projection: <https://www.youtube.com/watch?v=Vo9LC9d7FQA>

Unit V

Multidisciplinary Applications & Practice: (For CIE only)

Free hand Sketching: True free hand, Guided Free hand, Roads.

Drawing Simple Mechanisms: Bicycles, Tricycles, Gear trains.

Electric Wiring and lighting diagrams: Like, Automatic fire alarm using suitable software

Basic Building Drawing: Architectural floor plan, basic foundation drawing using suitable software.

Electronics Engineering Drawings: Simple Electronics Circuit Drawings, practice on layers concept

- Pedagogy/Course delivery tools: Chalk and talk, Power point presentation
- Links: Free hand sketching: <https://www.youtube.com/watch?v=Ess0dmJB2lo>
- Links: Electric wiring and lighting diagram: <https://www.youtube.com/watch?v=c67wIH2IJL8>
- Links: Electronics Engineering Drawings: <https://www.youtube.com/watch?v=RpF7oFC-LPY>

Course Outcomes (COs):

At the end of the course the student will be able to:

1. Demonstrate the usage of suitable software for creating basic and applied engineering drawings. (PO-1, PO-5, PO-9, PO-10, PO-12)
2. Conceptually sketch and draw developments for typical lateral surfaces. (PO-1, PO-5, PO-9, PO-10)
3. Exhibit the knowledge of orthographic and isometric projections of typical solids. (PO-1, PO-5, PO-9, PO-10)